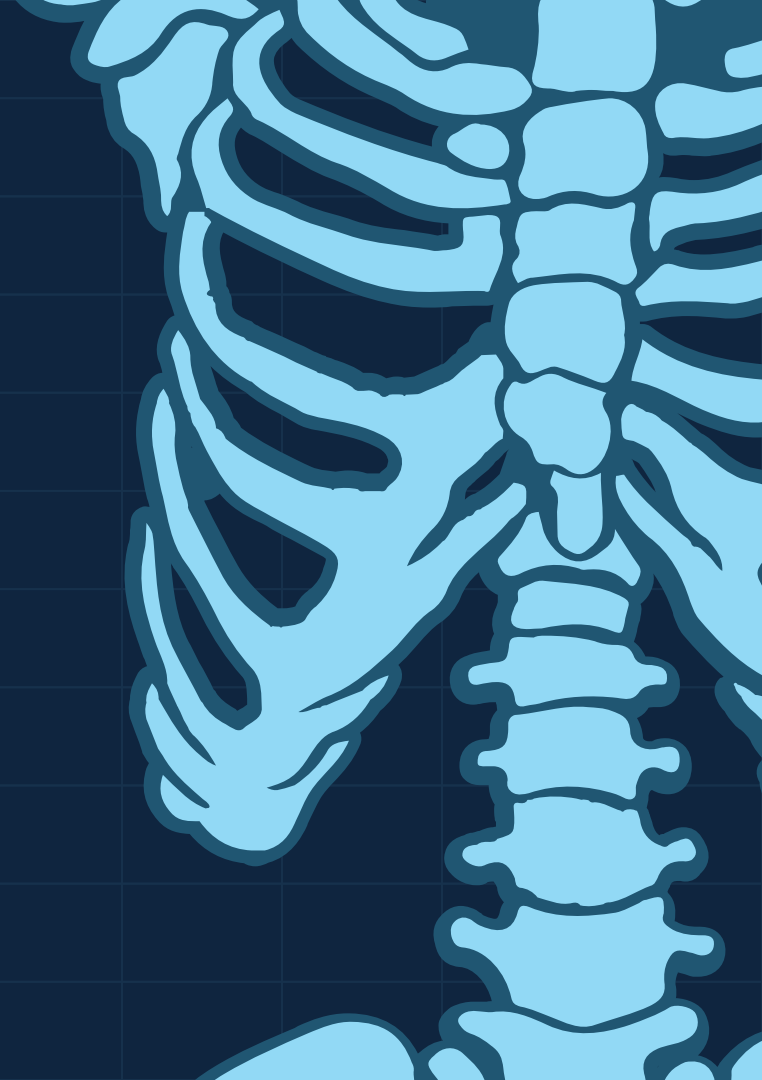
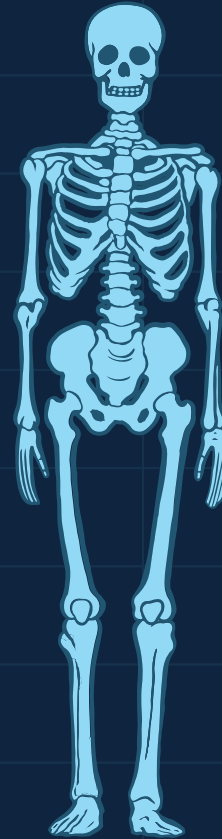


Catheter Placement

By Carlos, Emma, Fiona, Kevin, Pooja, Ryan

10cm





Background | Catheters

ETT



- Endotracheal tube
- Used to ventilate patient
- Remove blockages from airway

NGT



- Nasogastric tube
- Inserted nose to stomach
- Used for decompression of the stomach (intestinal obstruction)



CUC

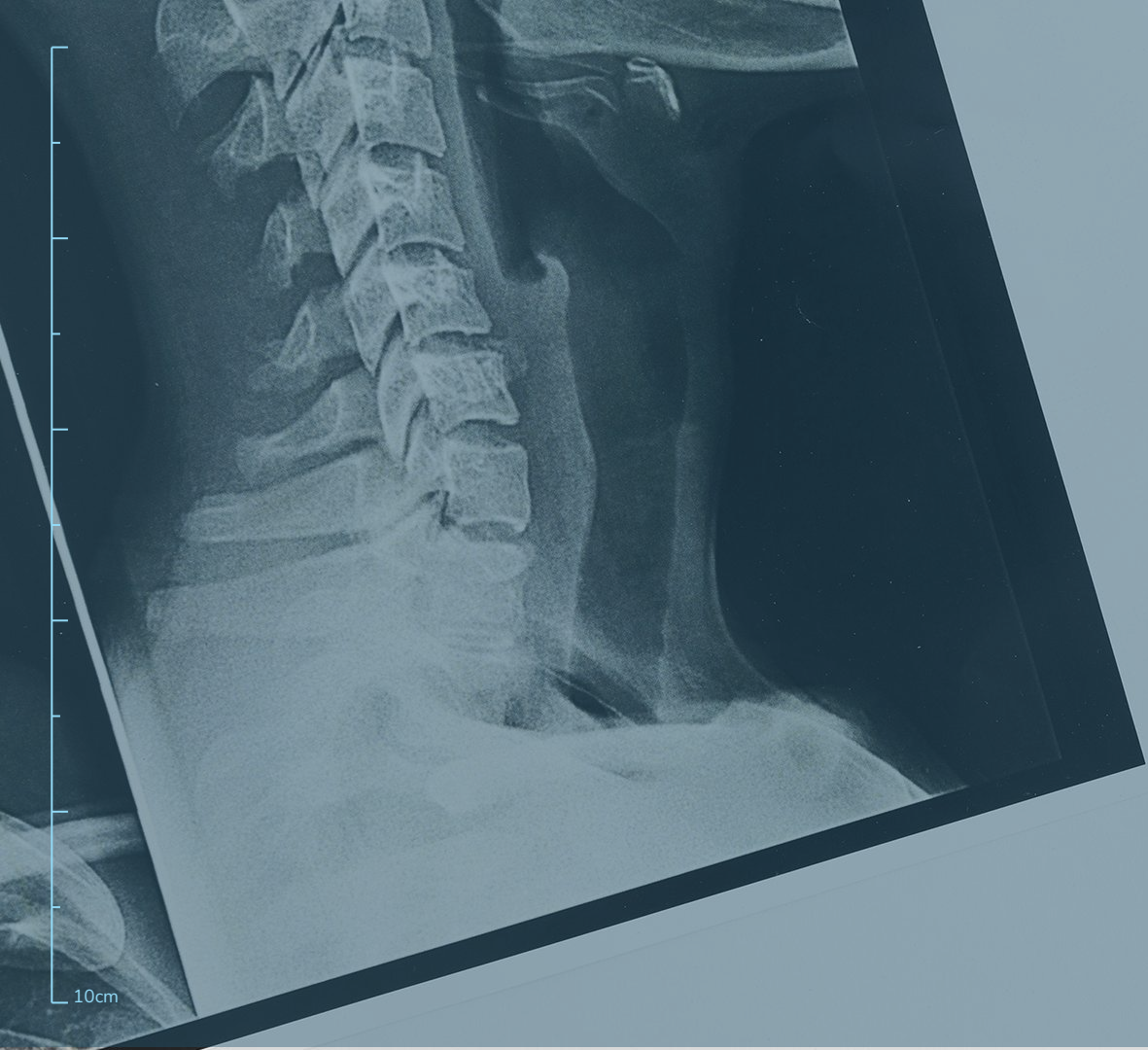


- Central venous catheter
- Receive drugs, fluids or blood for emergency or long-term treatment

Swan-Ganz



- Endotracheal tube
- Inserted into pulmonary artery
- Monitor heart condition and blood flow

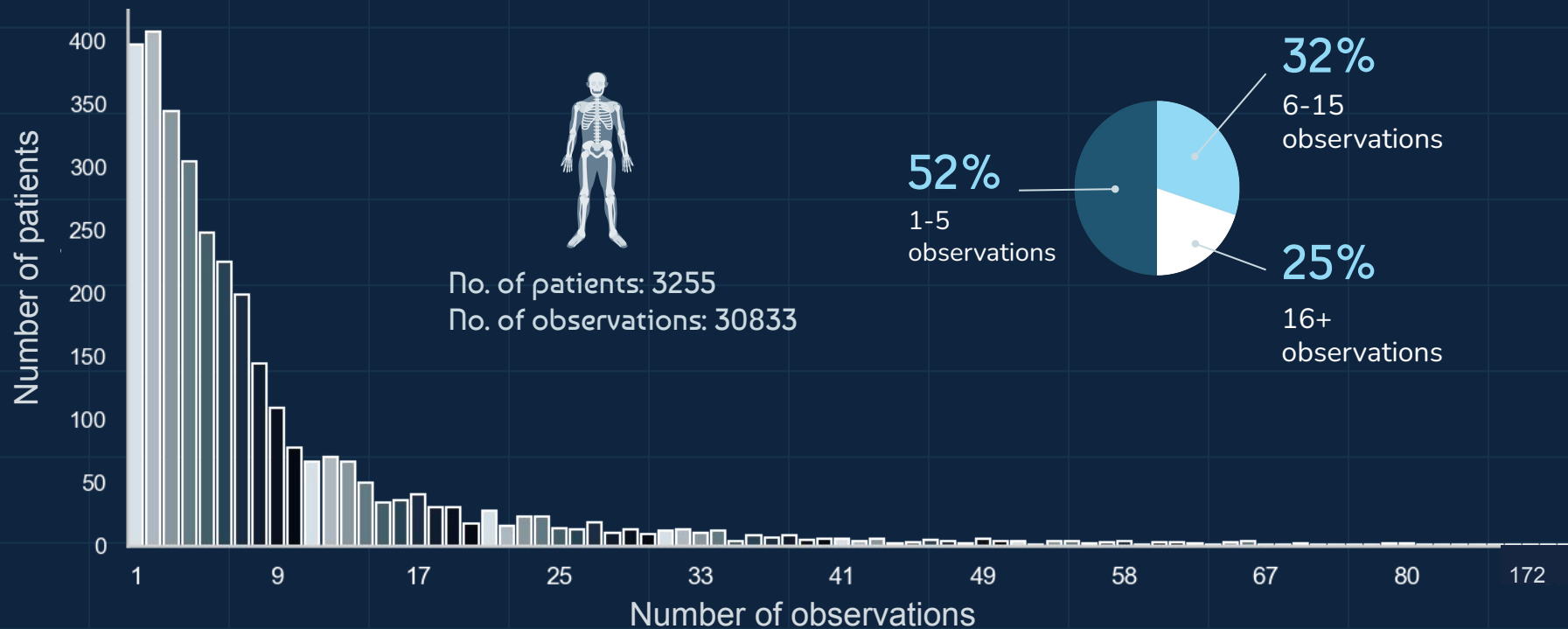
A lateral X-ray of a human cervical spine (neck). The vertebrae are clearly visible, and a 10cm scale bar is present in the bottom left corner. The image is tilted slightly to the right.

Research question

Which convolutional neural network out of ResNet, DenseNet, or UNet can accurately assess the placement accuracy of the ETT, NGT and CUC?

Exploratory Data Analysis

Exploratory Data Analysis



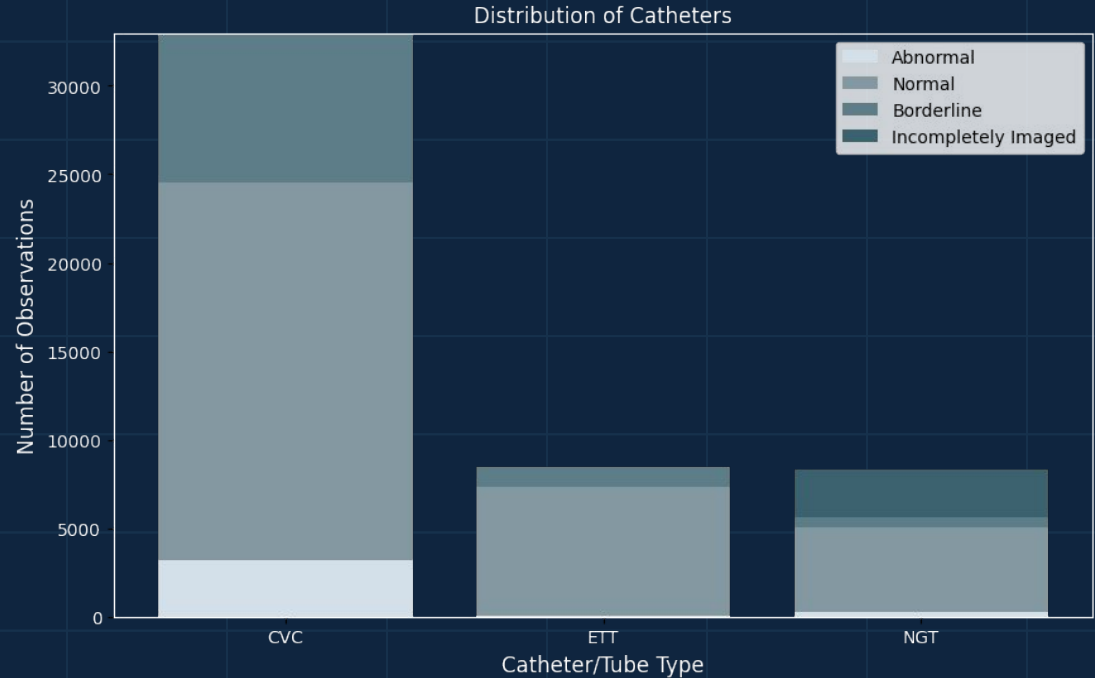
Exploratory Data Analysis

Normal: appropriately positioned

Borderline: ideally require some repositioning but would still function adequately

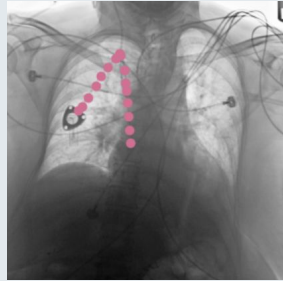
Abnormal: require immediate repositioning

Incompletely Imaged: catheter/tube position inconclusive due to imaging

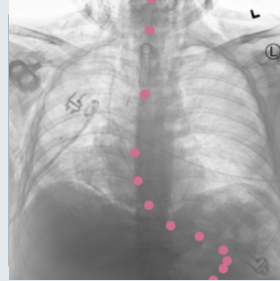


Exploratory Data Analysis

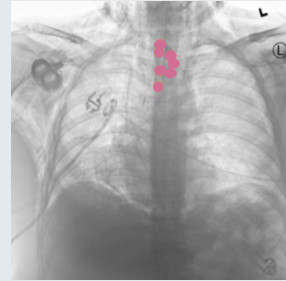
Normal



CUC

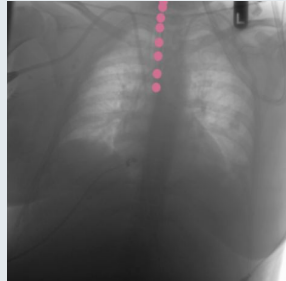
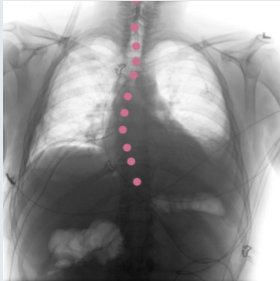
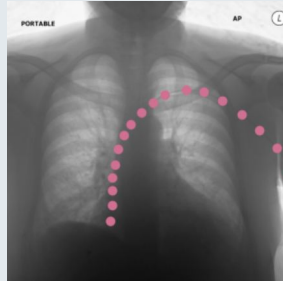


NGT



ETT

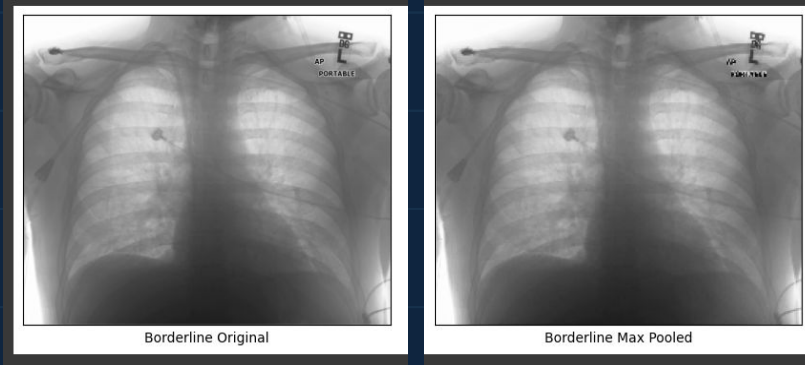
Abnormal



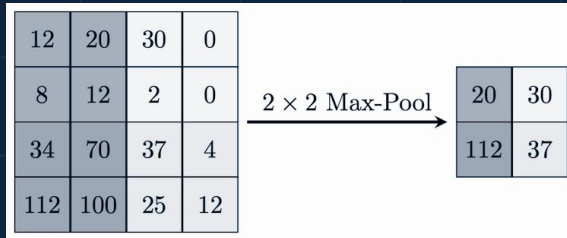
Data Modelling

Custom-built Neural Network, ResNet, DenseNet, UNet

Neural Network Preprocessing and Architecture



Utilised max pooling to downscale images

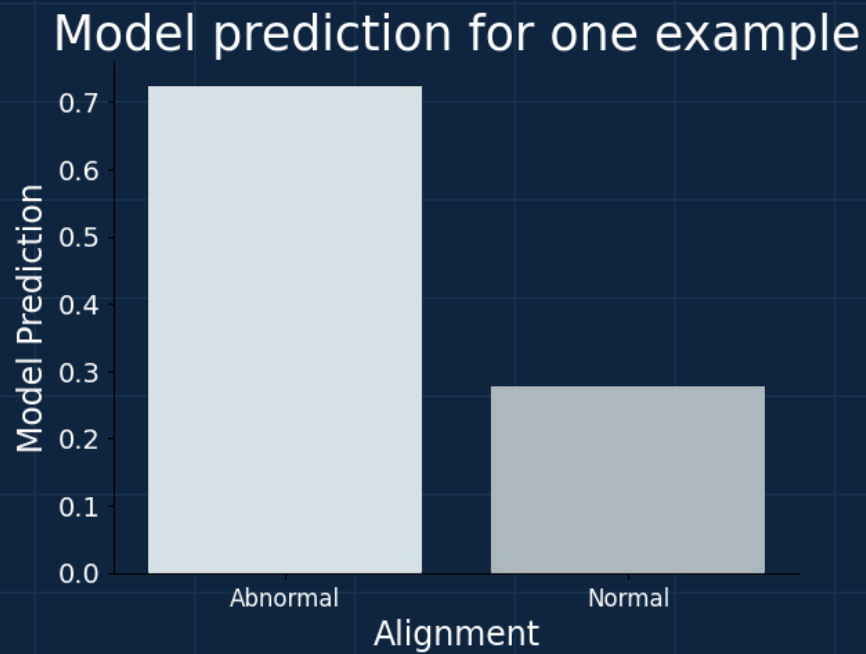
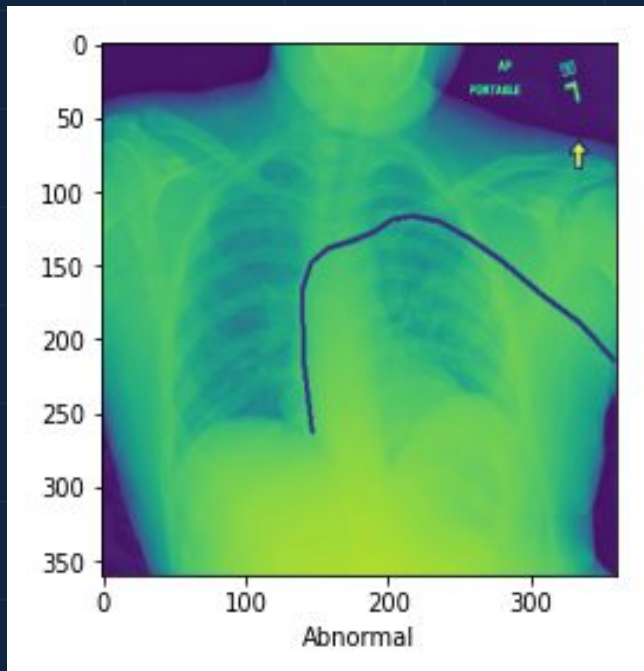


Architecture:

- 4 layers:
 - Input layer flattens image
 - 2 dense layers with ReLu activation
 - Classification layer has 1 node for the binary outputs

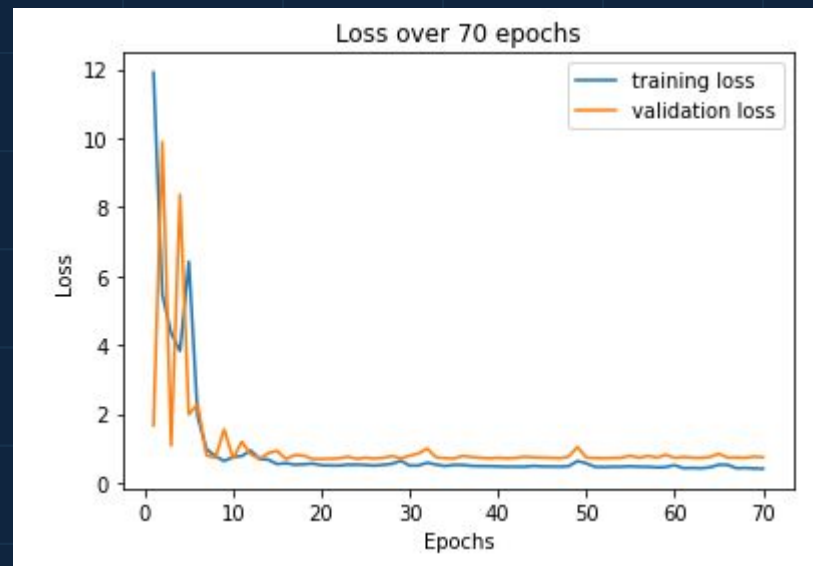
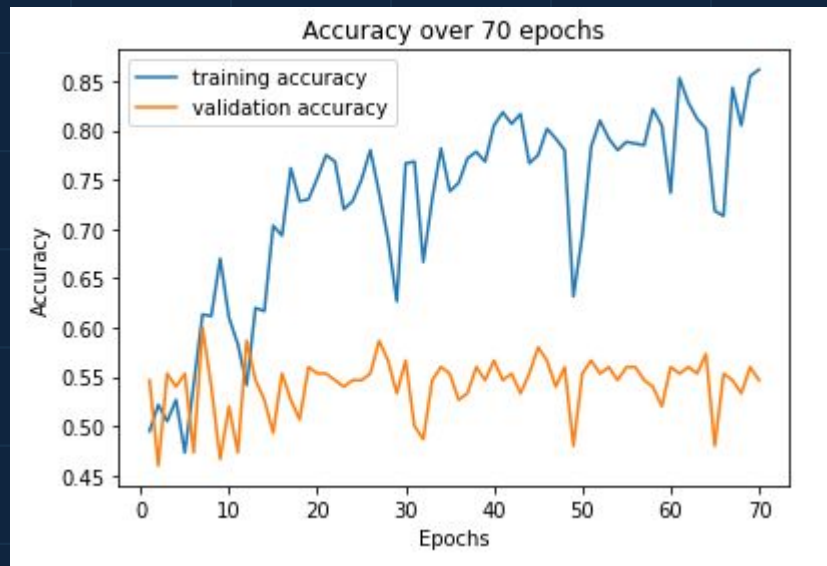
```
lr = 0.000005
epochs=70
img_size = (600, 600)
model = tf.keras.Sequential([
    tf.keras.layers.Flatten(input_shape=img_size),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(64, activation="relu"),
    tf.keras.layers.Dense(1)
])
```

Neural Network Isolated Example



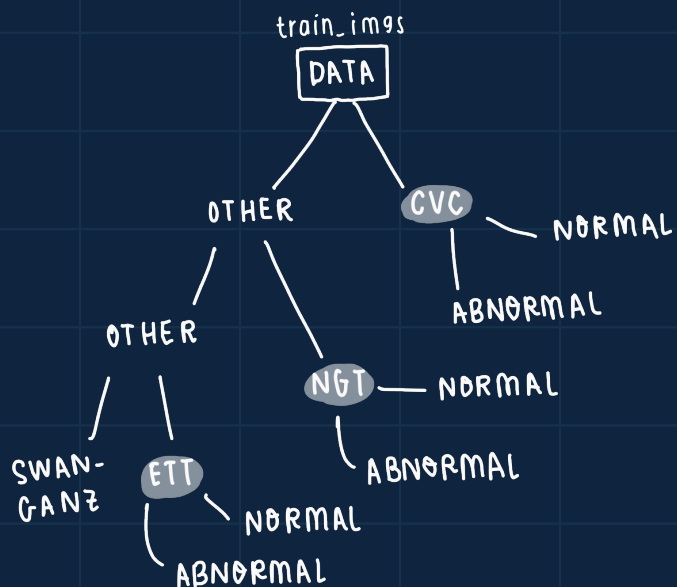
Binary Classification Neural Network Results

Training accuracy peaked at 0.84
Validation accuracy peaked at 0.60
Performs slightly better than a random guess



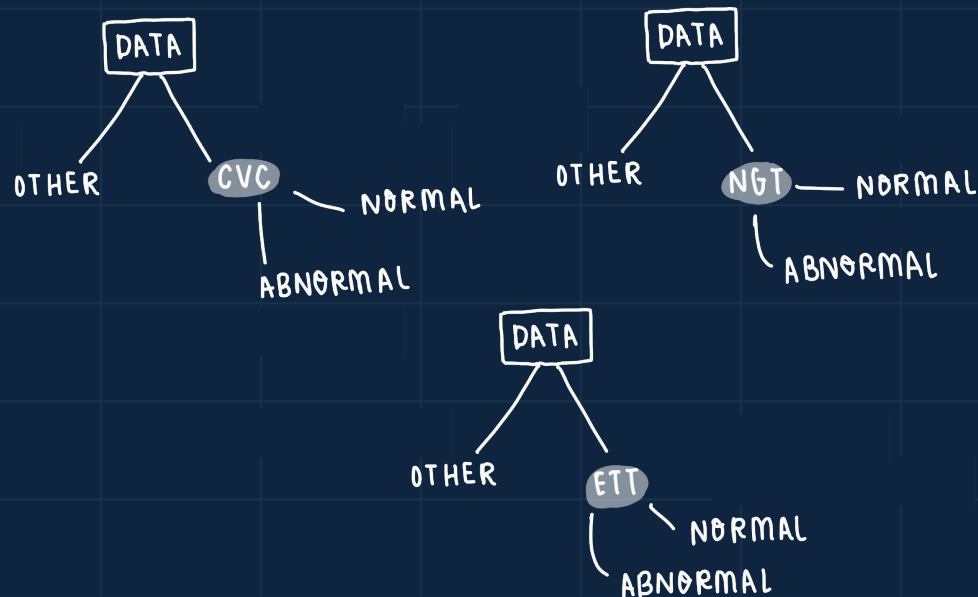
ResNet/DenseNet - Building our Models

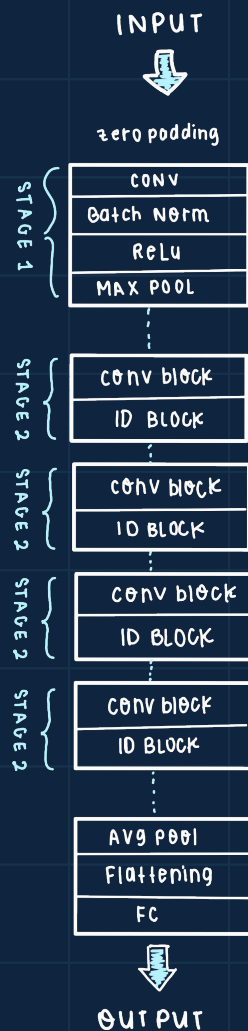
What we wanted to achieve:



What we achieved:

3 MODELS



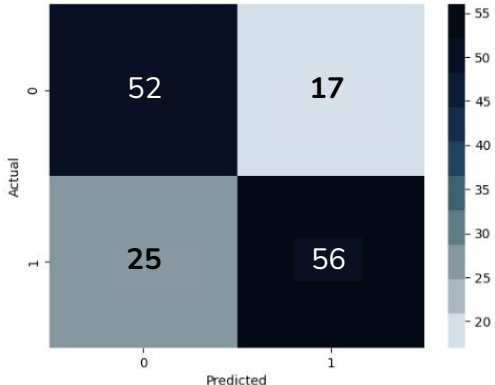
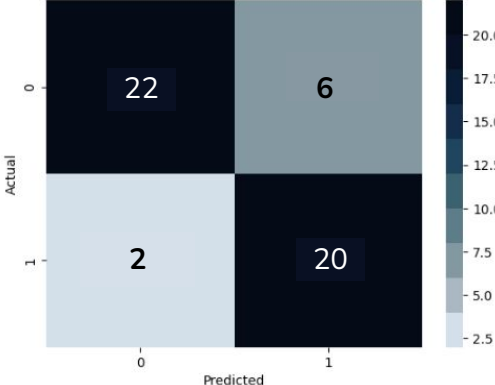
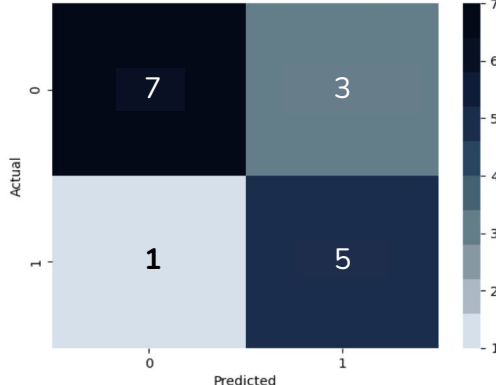


ResNet

- A pre-built convolution neural network used for image classification. A residual network.
- ResNet-50 used. A 50-layer CNN that uses a technique called residual learning.
 - Allows the model to learn more complex features by adding the output of each layer to the input of the next layer.

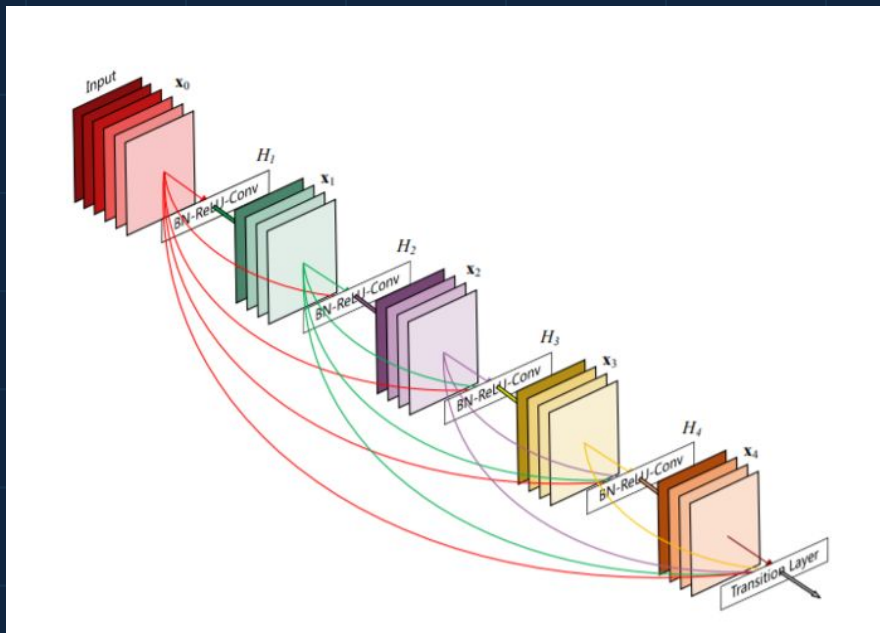


ResNet Models

Central Venous Catheter (CVC)	Nasogastric Tube (NGT)	Endotracheal Tube (ETT)																											
Training accuracy peaked at 0.99 Validation accuracy peaked at 0.78 Testing accuracy was 0.72	Training accuracy peaked at 0.99 Validation accuracy peaked at 0.84 Testing accuracy was 0.84	Training accuracy peaked at 0.99 Validation accuracy peaked at 0.81 Testing accuracy was 0.82																											
Confusion Matrix - Normal/Abnormal	Confusion Matrix - Normal/Abnormal	Confusion Matrix - Normal/Abnormal																											
 Confusion Matrix for CVC showing counts for Actual (0, 1) vs Predicted (0, 1). The matrix is: Actual 0: Predicted 0 = 52, Predicted 1 = 17; Actual 1: Predicted 0 = 25, Predicted 1 = 56. A color scale on the right ranges from 20 to 55. <table border="1"> <thead> <tr> <th>Actual \ Predicted</th><th>0</th><th>1</th></tr> </thead> <tbody> <tr> <th>0</th><td>52</td><td>17</td></tr> <tr> <th>1</th><td>25</td><td>56</td></tr> </tbody> </table>	Actual \ Predicted	0	1	0	52	17	1	25	56	 Confusion Matrix for NGT showing counts for Actual (0, 1) vs Predicted (0, 1). The matrix is: Actual 0: Predicted 0 = 22, Predicted 1 = 6; Actual 1: Predicted 0 = 2, Predicted 1 = 20. A color scale on the right ranges from 2.5 to 20.0. <table border="1"> <thead> <tr> <th>Actual \ Predicted</th><th>0</th><th>1</th></tr> </thead> <tbody> <tr> <th>0</th><td>22</td><td>6</td></tr> <tr> <th>1</th><td>2</td><td>20</td></tr> </tbody> </table>	Actual \ Predicted	0	1	0	22	6	1	2	20	 Confusion Matrix for ETT showing counts for Actual (0, 1) vs Predicted (0, 1). The matrix is: Actual 0: Predicted 0 = 7, Predicted 1 = 3; Actual 1: Predicted 0 = 1, Predicted 1 = 5. A color scale on the right ranges from 1 to 7. <table border="1"> <thead> <tr> <th>Actual \ Predicted</th><th>0</th><th>1</th></tr> </thead> <tbody> <tr> <th>0</th><td>7</td><td>3</td></tr> <tr> <th>1</th><td>1</td><td>5</td></tr> </tbody> </table>	Actual \ Predicted	0	1	0	7	3	1	1	5
Actual \ Predicted	0	1																											
0	52	17																											
1	25	56																											
Actual \ Predicted	0	1																											
0	22	6																											
1	2	20																											
Actual \ Predicted	0	1																											
0	7	3																											
1	1	5																											



DenseNet Model



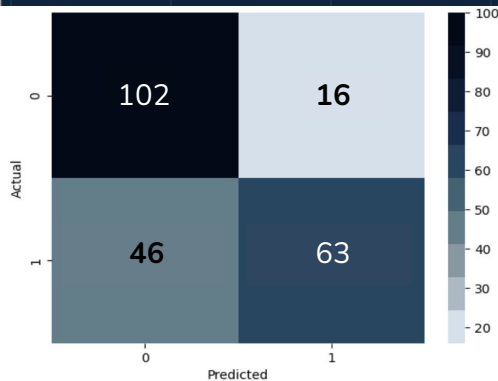
- ensuring that the feature-maps of the initial layers are used in all subsequent layers.
- better gradient flow,
- encourages feature reuse,
- reduces the risk of overfitting.

DenseNet Models

Central Venous Catheter (CVC)

Training accuracy peaked at 0.99
Validation accuracy peaked at 0.78
Testing accuracy was 0.72

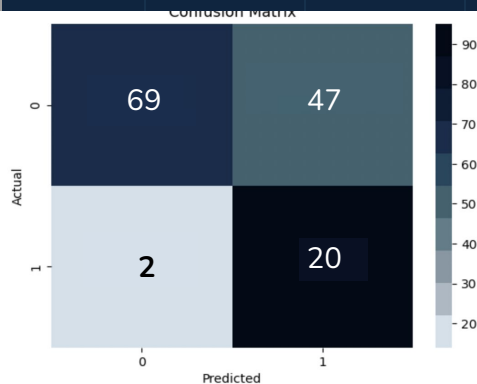
Confusion Matrix - Normal/Abnormal



Nasogastric Tube (NGT)

Training accuracy peaked at 0.99
Validation accuracy peaked at 0.74
Testing accuracy was 0.72

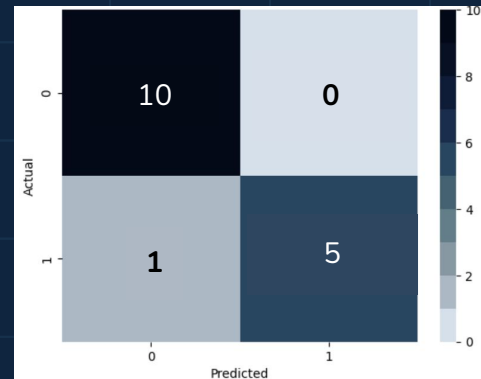
Confusion Matrix - Normal/Abnormal



Endotracheal Tube (ETT)

Training accuracy peaked at 0.99
Validation accuracy peaked at 0.88
Testing accuracy was 0.87

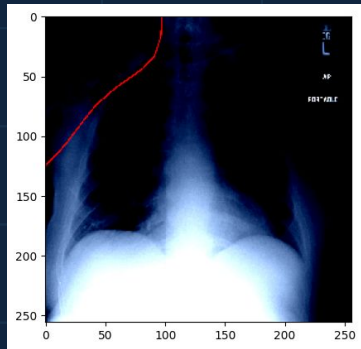
Confusion Matrix - Normal/Abnormal



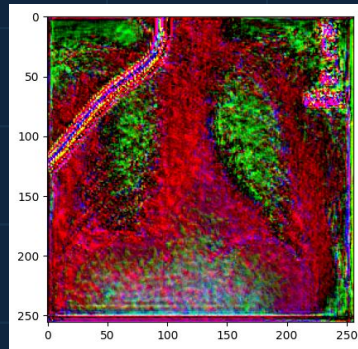
UNet Model

Convolutional neural network model designed for medical image segmentation.

- Composed of two main parts: Encoder and Decoder
- Pre-trained on ImageNet database.
- EfficientNet encoder (7 Million parameters)

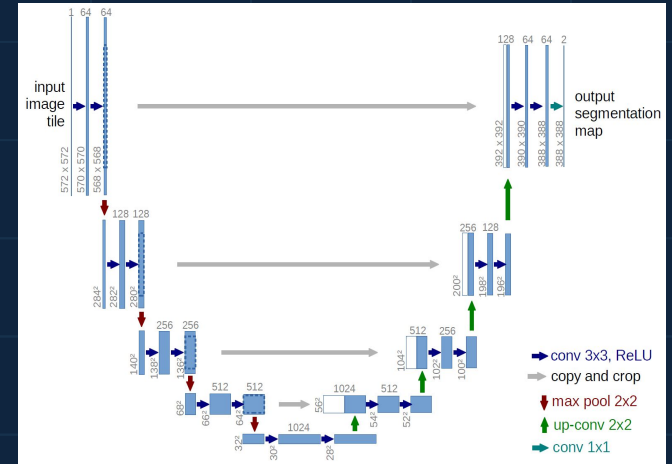


Example input image



Segmented image

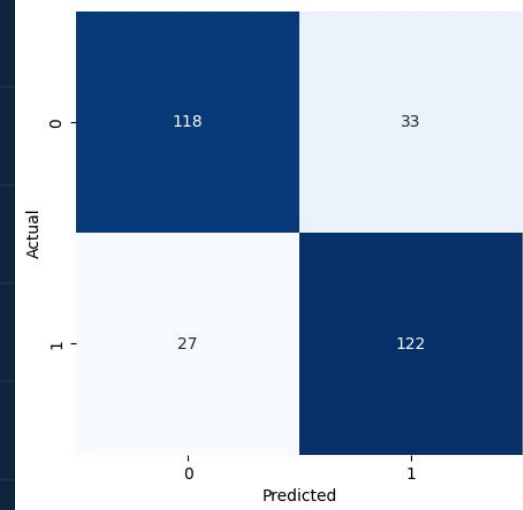
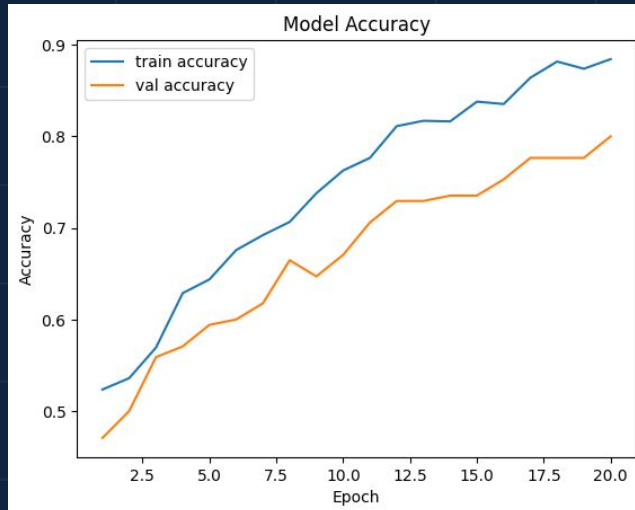
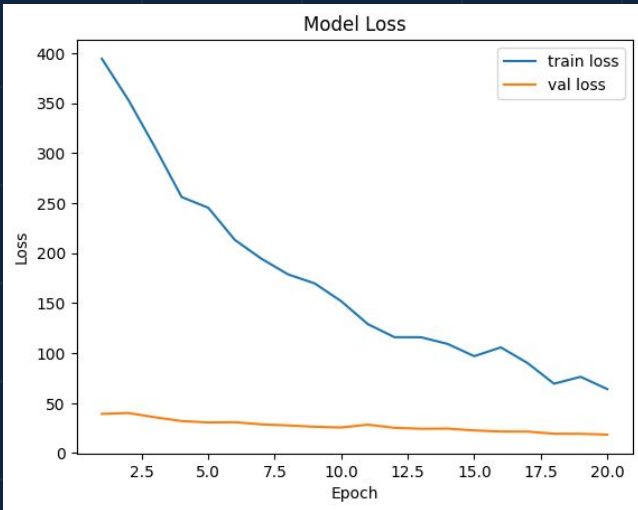
+ Image Classification



UNet Architecture

UNet Results

Training accuracy peaked at 0.83
Validation accuracy peaked at 0.80
Testing accuracy was 0.80



CUC Placement Prediction Accuracies

DEEP LEARNING IMAGE CLASSIFICATION MODELS



Custom-Built Model



55.40%



ResNet Model



72.00%



DenseNet Model



72.00%



UNet Model



80.00%

Thanks for listening

